Models 43 & 44 Precision Sub-Atmosphere Pressure Regulators

Features & Benefits

- Barometrically compensated to regulate absolute pressure ranges (Model 43)
- Used for compound (vacuum-pressure) ranges (Model 44)
- Precision bellows-sensing elements provide greater sensitivity

Description

The Models 43 and 44 Sub-Atmospheric Pressure Regulators are pilot-operated regulators with bellows-sensing elements for use within absolute and compound pressure ranges. The Model 43 is barometrically compensated for use in absolute pressure ranges, while the Model 44 regulates compound (vacuum-pressure) ranges.

Both models use the pneumatic null-balance principle of operation. As such, imbalances between the regulator setting and the output are corrected by a pilot circuit that holds the output pressure constant regardless of wide changes in flow or supply pressure.

A vacuum pump provides the sub-atmospheric pressures. In addition, both model series have tapped exhaust connections for use with this pump.

Operating Characteristics

Standard Supply Pressure

At least 5 psi higher than the maximum desired regulated pressure

Maximum Supply Pressure

Model 43-20: 20 psig
Model 43-50: 55 psig
Model 44-20: 35 psig
Model 44-50: 70 psig

Minimum Supply Pressure

3 psig above regulated output

Overrange Protection

The application of the maximum rated supply pressure to any connection will not damage the instrument

Ambient Temperature Limits

-40 to 180°F (-40 to 80°C)

Specifications

Resolution Adjustment

Within ± 0.001 Hg of desired pressure setting

Supply Pressure

Minimum: 3 psig above regulated output



Ambient Temperature Effect (for 50°F [27°C] temperature change)

Model 43-20 & 43-50: ±1% of regulated pressure
Model 44-20 & 44-50: ±0.1 psi
Model 43L: ±0.03 psi

Pilot-air Consumption (with recommended supply)

Model 43-20 & 44-20: 0.06 scfm
Model 43-50 & 44-50: 0.10 scfm
For L-type instrument, in balanced condition on deadend service: scfm - 0.003 $\sqrt{P}$ where P = set pressure in psia

Standard Mounting

In-line pipe or flush panel up to 3/16" thick
(1-3/32 dia. hole req'd for panel mounting)

Connections

1/4" NPT for supply & outlet, all other 1/8" NPT

Materials of Construction

Brass, stainless steel, Neoprene, aluminum, and zinc

Model No.	Range	Regulator Body Material
43-20	1-20 psia	brass
43-50	1-50 psia	brass
43-20L[1]	0.1" Hg-20 psia	brass
43-50L[1]	0.1" Hg 50 psia	brass
43A20L[1]	0.1" Hg-20 psia	aluminum
43A50L[1]	0.1" Hg 50 psia	aluminum
44-20	Nominal 28" Hg Vac 5 psig	brass
44-50	Nominal 28" Hg Vac 35 psig	brass

1) A constant differential relay is required with these units in order to reach the low pressures listed. For a piped assembly (absolute pressure regulator with constant differential relay), add 63A to the basic model number (i.e. -43-20L would be designated -43-20L63A).

Dimensional drawings

Mounting Dimensions

